

Example 4.18

Find the value of

(i) $\sin \left[\frac{\pi}{3} - \sin^{-1} \left(-\frac{1}{2} \right) \right]$

(ii) $\cos \left[\frac{1}{2} \cos^{-1} \left(\frac{1}{8} \right) \right]$

(iii) $\tan \left[\frac{1}{2} \sin^{-1} \left(\frac{2a}{1+a^2} \right) + \frac{1}{2} \cos^{-1} \left(\frac{1-a^2}{1+a^2} \right) \right]$.

Solution

(i) $\sin \left[\frac{\pi}{3} - \sin^{-1} \left(-\frac{1}{2} \right) \right] = \sin \left[\frac{\pi}{3} - \left(-\frac{\pi}{6} \right) \right] = \sin \left(\frac{\pi}{2} \right) = 1.$

(ii) Consider $\cos \left[\frac{1}{2} \cos^{-1} \left(\frac{1}{8} \right) \right]$. Let $\cos^{-1} \left(\frac{1}{8} \right) = \theta$. Then, $\cos \theta = \frac{1}{8}$ and $\theta \in [0, \pi]$.

Now, $\cos \theta = \frac{1}{8}$ implies $2 \cos^2 \frac{\theta}{2} - 1 = \frac{1}{8}$. Thus, $\cos \left(\frac{\theta}{2} \right) = \frac{3}{4}$, since $\cos \left(\frac{\theta}{2} \right)$ is positive.

Thus, $\cos \left[\frac{1}{2} \cos^{-1} \left(\frac{1}{8} \right) \right] = \cos \left(\frac{\theta}{2} \right) = \frac{3}{4}.$

(iii) $\tan \left[\frac{1}{2} \sin^{-1} \left(\frac{2a}{1+a^2} \right) + \frac{1}{2} \cos^{-1} \left(\frac{1-a^2}{1+a^2} \right) \right]$

Let $a = \tan \theta$.

Now,

$$\begin{aligned} \tan \left[\frac{1}{2} \sin^{-1} \left(\frac{2a}{1+a^2} \right) + \frac{1}{2} \cos^{-1} \left(\frac{1-a^2}{1+a^2} \right) \right] &= \tan \left[\frac{1}{2} \sin^{-1} \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right) + \frac{1}{2} \cos^{-1} \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right) \right] \\ &= \tan \left[\frac{1}{2} \sin^{-1} (\sin 2\theta) + \frac{1}{2} \cos^{-1} (\cos 2\theta) \right] = \tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{2a}{1 - a^2}. \end{aligned}$$

Example 4.19

Prove that $\tan(\cos^{-1} x) = \frac{\sqrt{1-x^2}}{x}$ for $|x| < 1, x \neq 0$.

Solution

Let $\cos^{-1} x = \theta$. Then, $x = \cos \theta$ and $-1 \leq x \leq 1$.

$$\text{Now, } \tan(\cos^{-1} x) = \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sqrt{1 - \cos^2 \theta}}{\cos \theta} = \frac{\sqrt{1 - x^2}}{x}, \quad |x| < 1, x \neq 0.$$

Example 4.20

Evaluate $\sin \left[\sin^{-1} \left(\frac{3}{5} \right) + \sec^{-1} \left(\frac{5}{4} \right) \right]$

Solution

Let $\sec^{-1} \frac{5}{4} = \theta$. Then, $\sec \theta = \frac{5}{4}$ and hence, $\cos \theta = \frac{4}{5}$.

Also, $\sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - \left(\frac{4}{5} \right)^2} = \frac{3}{5}$, which gives $\theta = \sin^{-1} \left(\frac{3}{5} \right)$.

Thus, $\sec^{-1} \left(\frac{5}{4} \right) = \sin^{-1} \left(\frac{3}{5} \right)$ and $\sin^{-1} \frac{3}{5} + \sec^{-1} \left(\frac{5}{4} \right) = 2 \sin^{-1} \left(\frac{3}{5} \right)$.

We know that $\sin^{-1} \left(2x\sqrt{1-x^2} \right) = 2 \sin^{-1} x$, if $|x| \leq \frac{1}{\sqrt{2}}$.

Since $\frac{3}{5} < \frac{1}{\sqrt{2}}$, we have $2 \sin^{-1} \left(\frac{3}{5} \right) = \sin^{-1} \left(2 \times \frac{3}{5} \sqrt{1 - \left(\frac{3}{5} \right)^2} \right) = \sin^{-1} \left(\frac{24}{25} \right)$.

Hence, $\sin \left[\sin^{-1} \left(\frac{3}{5} \right) + \sec^{-1} \left(\frac{5}{4} \right) \right] = \sin \left(\sin^{-1} \left(\frac{24}{25} \right) \right) = \frac{24}{25}$, since $\frac{24}{25} \in [-1, 1]$.

Example 4.21

Prove that (i) $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{4}$ (ii) $2 \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{7} = \tan^{-1} \frac{31}{17}$

Solution

(i) We know that $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$, $xy < 1$.

$$\text{Thus, } \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \tan^{-1} \frac{\frac{1}{2} + \frac{1}{3}}{1 - \left(\frac{1}{2}\right)\left(\frac{1}{3}\right)} = \tan^{-1}(1) = \frac{\pi}{4}.$$

(ii) We know that $2 \tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2}$, $-1 < x < 1$.

$$\text{So, } 2 \tan^{-1} \frac{1}{2} = \tan^{-1} \frac{2\left(\frac{1}{2}\right)}{1 - \left(\frac{1}{2}\right)^2} = \tan^{-1} \left(\frac{4}{3}\right).$$

$$\text{Hence, } 2 \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{7} = \tan^{-1} \frac{4}{3} + \tan^{-1} \frac{1}{7} = \tan^{-1} \left(\frac{\frac{4}{3} + \frac{1}{7}}{1 - \left(\frac{4}{3}\right)\left(\frac{1}{7}\right)} \right) = \tan^{-1} \left(\frac{31}{17} \right).$$

Example 4.22

If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi$ and $0 < x, y, z < 1$, show that

$$x^2 + y^2 + z^2 + 2xyz = 1$$

Solution

Let $\cos^{-1} x = \alpha$ and $\cos^{-1} y = \beta$. Then, $x = \cos \alpha$ and $y = \cos \beta$.

$\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi$ gives $\alpha + \beta = \pi - \cos^{-1} z$.

Now, $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta = xy - \sqrt{1-x^2} \sqrt{1-y^2}$.

From (1), we get $\cos(\pi - \cos^{-1} z) = xy - \sqrt{1-x^2} \sqrt{1-y^2}$

$$-\cos(\cos^{-1} z) = xy - \sqrt{1-x^2} \sqrt{1-y^2}.$$

So, $-z = xy - \sqrt{1-x^2} \sqrt{1-y^2}$, which gives $-xy - z = -\sqrt{1-x^2} \sqrt{1-y^2}$.

Squaring on both sides and simplifying, we get $x^2 + y^2 + z^2 + 2xyz = 1$.

Example 4.23

If $a_1, a_2, a_3, \dots, a_n$ is an arithmetic progression with common difference d ,

prove that
$$\tan \left[\tan^{-1} \left(\frac{d}{1 + a_1 a_2} \right) + \tan^{-1} \left(\frac{d}{1 + a_2 a_3} \right) + \dots + \tan^{-1} \left(\frac{d}{1 + a_n a_{n-1}} \right) \right] = \frac{a_n - a_1}{1 + a_1 a_n}.$$

Solution

Now,
$$\tan^{-1} \left(\frac{d}{1 + a_1 a_2} \right) = \tan^{-1} \left(\frac{a_2 - a_1}{1 + a_1 a_2} \right) = \tan^{-1} a_2 - \tan^{-1} a_1.$$

Similarly,
$$\tan^{-1} \left(\frac{d}{1 + a_2 a_3} \right) = \tan^{-1} \left(\frac{a_3 - a_2}{1 + a_2 a_3} \right) = \tan^{-1} a_3 - \tan^{-1} a_2.$$

Continuing inductively, we get

$$\tan^{-1} \left(\frac{d}{1 + a_n a_{n-1}} \right) = \tan^{-1} \left(\frac{a_n - a_{n-1}}{1 + a_{n-1} a_n} \right) = \tan^{-1} a_n - \tan^{-1} a_{n-1}.$$

Adding vertically, we get

$$\begin{aligned} \tan^{-1} \left(\frac{d}{1 + a_1 a_2} \right) + \tan^{-1} \left(\frac{d}{1 + a_2 a_3} \right) + \dots + \tan^{-1} \left(\frac{d}{1 + a_n a_{n-1}} \right) &= \tan^{-1} a_n - \tan^{-1} a_1. \\ \tan \left[\tan^{-1} \left(\frac{d}{1 + a_1 a_2} \right) + \tan^{-1} \left(\frac{d}{1 + a_2 a_3} \right) + \dots + \tan^{-1} \left(\frac{d}{1 + a_n a_{n-1}} \right) \right] &= \tan [\tan^{-1} a_n - \tan^{-1} a_1] \\ &= \tan \left[\tan^{-1} \left(\frac{a_n - a_1}{1 + a_1 a_n} \right) \right] = \frac{a_n - a_1}{1 + a_1 a_n}. \end{aligned}$$

Example 4.24 Solve $\tan^{-1} \left(\frac{1-x}{1+x} \right) = \frac{1}{2} \tan^{-1} x$ for $x > 0$.

Solution

$$\tan^{-1} \left(\frac{1-x}{1+x} \right) = \frac{1}{2} \tan^{-1} x \quad \text{gives} \quad \tan^{-1} (1 - \tan^2 x) = \frac{1}{2} \tan^{-1} x.$$

Therefore, $\frac{x}{\sqrt{2}} = \frac{1}{2} \tan^{-1} x$, which in turn reduces to $\tan^{-1} x = \frac{x}{6}$.

Thus, $x = \tan \left(\frac{x}{6} \right) = \frac{1}{\sqrt{3}}$.

Example 4.25 Solve $\sin^{-1} x > \cos^{-1} x$

Solution

Given that $\sin^{-1} x > \cos^{-1} x$. Note that $-1 \leq x \leq 1$.

Adding both sides by $\sin^{-1} x$, we get

$$\sin^{-1} x + \sin^{-1} x > \cos^{-1} x + \sin^{-1} x, \text{ which reduces to } 2 \sin^{-1} x > \frac{\pi}{2}.$$

As sine function increases in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, we have $x > \sin \frac{\pi}{4}$ or $x > \frac{1}{\sqrt{2}}$.

Thus, the solution set is the interval $\left(\frac{1}{\sqrt{2}}, 1\right]$.

Example 4.26

Show that $\cos(\sin^{-1} x) = \frac{\sqrt{1-x^2}}{x}$, $-1 \leq x \leq 1$ and $x \neq 0$

Solution

Let $\sin^{-1} x = \theta$. Then, $x = \sin \theta$ and $x \neq 0$, we get $\theta \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$.

Hence, $\cos \theta > 0$ and $\cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - x^2}$.

Thus, $\cos(\sin^{-1} x) = \cos \theta = \frac{\sqrt{1-x^2}}{x}$, $-1 \leq x \leq 1$ and $x \neq 0$

Example 4.27: Solve $\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4}$, if $6x^2 < 1$.

Solution

Now, $\tan^{-1} 2x + \tan^{-1} 3x = \tan^{-1} \left(\frac{2x + 3x}{1 - 6x^2} \right)$, since $6x^2 < 1$.

So, $\tan^{-1} \left(\frac{5x}{1 - 6x^2} \right) = \frac{\pi}{4}$, which implies $\frac{5x}{1 - 6x^2} = \tan \frac{\pi}{4} = 1$.

Thus, $1 - 6x^2 = 5x$, which gives $6x^2 + 5x - 1 = 0$

Hence, $x = \frac{1}{6}, -1$. But $x = -1$ does not satisfy $6x^2 < 1$.

Observe that $x = -1$ makes the left side of the equation negative whereas the right side is a positive number. Thus, $x = -1$ is not a solution. Hence, $x = \frac{1}{6}$ is the only solution of the equation.

Example 4.28

Solve $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$.

Solution

Now, $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \tan^{-1}\left[\frac{\frac{x-1}{x-2} + \frac{x+1}{x+2}}{1 - \frac{x-1}{x-2} \cdot \frac{x+1}{x+2}}\right] = \frac{\pi}{4}$.

Thus, $\frac{\frac{x-1}{x-2} + \frac{x+1}{x+2}}{1 - \frac{x-1}{x-2} \cdot \frac{x+1}{x+2}} = 1$, which on simplification gives $2x^2 - 4x = -3$

Thus, $x^2 - \frac{1}{2}$ gives $x = -\frac{1}{\sqrt{2}}$.

Example 4.29

Solve $\cos \left(\sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right) \right) = \sin \left\{ \cot^{-1} \left(\frac{3}{4} \right) \right\}.$

Solution

We know that $\sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right) = \cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right).$

Thus, $\cos \left(\sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right) \right) = \frac{1}{\sqrt{1+x^2}}.$

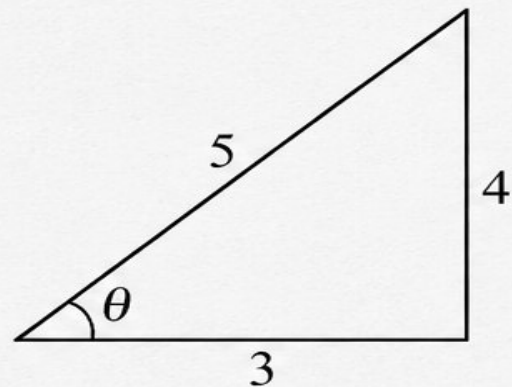
Let $\cot^{-1} \left(\frac{3}{4} \right) = \theta$. Then $\cot \theta = \frac{3}{4}$, and so θ is acute.

From the diagram, we get

Hence, $\sin \left\{ \cot^{-1} \left(\frac{3}{4} \right) \right\} = \sin \theta = \frac{4}{5}.$... (2)

Using (1) and (2) in the given equation, we get $\frac{1}{\sqrt{1+x^2}} = \frac{4}{5}$, which gives $\sqrt{1+x^2} = \frac{5}{4}.$

Thus, $x = \pm \frac{3}{4}.$



... (1)

Question 1.

Find the value, if it exists. If not, give the reason for non-existence.

(i) $\sin^{-1}(\cos \pi)$

(ii) $\tan^{-1}\left(\sin\left(-\frac{5\pi}{2}\right)\right)$

(iii) $\sin^{-1}[\sin 5]$

Solution:

(i) $\sin^{-1}(\cos \pi) = \sin^{-1}(-1) = -\frac{\pi}{2}$

(ii) $\tan^{-1}\left(\sin \frac{5\pi}{2}\right) = \tan^{-1}\left(-\sin \frac{\pi}{2}\right) = \tan^{-1}(-1) = -\frac{\pi}{4}$

(iii) $\sin^{-1}(\sin 5) = \sin^{-1}[\sin (5 - 2\pi)] = 5 - 2\pi$

Question 2.

Find the value of the expression in terms of x , with the help of a reference triangle.

- (i) $\sin(\cos^{-1}(1-x))$
- (ii) $\cos(\tan^{-1}(3x-1))$
- (iii) $\tan(\sin^{-1}(x + \frac{1}{2}))$

Solution:

(i) Let $\cos^{-1}(1-x) = \theta$

$$1-x = \cos \theta$$

We know $\sin^2 \theta = 1 - \cos^2 \theta$

$$\begin{aligned}\Rightarrow \sin \theta &= \sqrt{1 - \cos^2 \theta} = \sqrt{1 - (1-x)^2} = \sqrt{1 - (1+x^2-2x)} \\ &= \sqrt{2x - x^2}\end{aligned}$$

$$\Rightarrow \sin[\cos^{-1}(1-x)] = \sin \theta = \sqrt{2x - x^2}$$

(iii) Let $\sin^{-1}(x + \frac{1}{2}) = \theta \quad \Rightarrow \quad x + \frac{1}{2} = \sin \theta$

$$\Rightarrow 2x + 1 = 2 \sin \theta \quad \Rightarrow \quad \sin \theta = \frac{2x+1}{2}$$

$$\begin{aligned}\cos \theta &= \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \left(\frac{2x+1}{2}\right)^2} \\ &= \sqrt{1 - \frac{(4x^2+4x+1)}{4}} = \sqrt{\frac{4-4x^2-4x-1}{4}} = \frac{1}{2} \sqrt{3-4x-4x^2}\end{aligned}$$

$$\text{Now } \tan(\theta) = \frac{\sin \theta}{\cos \theta} = \frac{\frac{2x+1}{2}}{\frac{1}{2} \sqrt{3-4x-4x^2}} = \frac{2x+1}{\sqrt{3-4x-4x^2}} = \tan(\sin^{-1}(x + \frac{1}{2}))$$

(ii) Let $\tan^{-1}(3x-1) = \theta \quad \Rightarrow \quad 3x-1 = \tan \theta$

$$\text{So } \tan^2 \theta = (3x-1)^2 = 9x^2 - 6x + 1$$

We know $1 + \tan^2 \theta = \sec^2 \theta = \frac{1}{\cos^2 \theta}$

$$\Rightarrow \frac{1}{\cos^2 \theta} = 9x^2 - 6x + 1 + 1 = 9x^2 - 6x + 2$$

$$\text{So } \cos^2 \theta = \frac{1}{9x^2 - 6x + 2} \quad \Rightarrow \quad \cos \theta = \frac{1}{\sqrt{9x^2 - 6x + 2}}$$

$$\therefore \cos[\tan^{-1}(3x-1)] = \frac{1}{\sqrt{9x^2 - 6x + 2}}$$

Question 3.

Find the value of

(i) $\sin^{-1} \left(\cos \left(\sin^{-1} \frac{\sqrt{3}}{2} \right) \right)$

(ii) $\cot \left(\sin^{-1} \frac{3}{5} + \sin^{-1} \frac{4}{5} \right)$

(iii) $\tan \left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2} \right)$

Solution:

(i) $\sin^{-1} \frac{\sqrt{3}}{2} = \frac{\pi}{3}$ and $\cos \left(\frac{\pi}{3} \right) = \frac{1}{2}$ and

so $\sin^{-1} \left[\cos \left\{ \sin^{-1} \left(\frac{\sqrt{3}}{2} \right) \right\} \right] = \sin^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{6}$

(ii) Let $\sin^{-1} \frac{4}{5} = \theta$

$\Rightarrow \sin \theta = \frac{4}{5}$

$\Rightarrow \cos \theta = \frac{3}{5} \quad \Rightarrow \quad \theta = \cos^{-1} \frac{3}{5}$

$\therefore \cot \left[\sin^{-1} \frac{3}{5} + \sin^{-1} \frac{4}{5} \right] = \cot \left[\sin^{-1} \frac{3}{5} + \cos^{-1} \frac{3}{5} \right]$
 $= \cot \left(\frac{\pi}{2} \right) = 0$

(iii) Let $\sin^{-1} \frac{3}{5} = \theta$

$\Rightarrow \sin \theta = \frac{3}{5}$

$\Rightarrow \tan \theta = \frac{4}{4}$

so $\theta = \tan^{-1} \frac{3}{4}$

so $\theta = \tan^{-1} \frac{3}{4}$

Now let $\cot^{-1} \frac{3}{2} = \theta$

$\Rightarrow \cot \theta = \frac{3}{2}$

$\Rightarrow \tan \theta = \frac{2}{3} \quad \Rightarrow \quad \theta = \tan^{-1} \frac{2}{3}$

Now $\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2} = \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{2}{3}$

$= \tan^{-1} \left(\frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4} \times \frac{2}{3}} \right)$

$= \tan^{-1} \left(\frac{\frac{9+8}{12}}{1 - \frac{6}{12}} \right) = \tan^{-1} \left(\frac{17}{6} \right)$

so $\tan \left[\tan^{-1} \left(\frac{17}{6} \right) \right] = \frac{17}{6}$

Question 4.

Prove that

$$(i) \tan^{-1} \frac{2}{11} + \tan^{-1} \frac{7}{24} = \tan^{-1} \frac{1}{2}$$

$$(ii) \sin^{-1} \frac{3}{5} - \cos^{-1} \frac{12}{13} = \sin^{-1} \frac{18}{65}$$

Solution:

$$\begin{aligned}(i) \quad \tan^{-1} \frac{2}{11} + \tan^{-1} \frac{7}{24} &= \tan^{-1} \left[\frac{\left(\frac{2}{11}\right) + \left(\frac{7}{24}\right)}{1 - \frac{2}{11} \times \frac{7}{24}} \right] = \tan^{-1} \left[\frac{\frac{48+77}{11 \times 24}}{\frac{14}{11 \times 24}} \right] \\ &= \tan^{-1} \left[\frac{125}{14} \right] = \tan^{-1} \left[\frac{1}{2} \right] \quad \left(\because \frac{125}{14} = \frac{1}{2} \text{ when reduced} \right)\end{aligned}$$

$$(ii) \quad \text{Let } \cos^{-1} \frac{12}{13} = \theta \quad \Rightarrow \quad \cos \theta = \frac{12}{13}$$

$$\text{so} \quad \sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - \frac{144}{169}} = \sqrt{\frac{25}{169}} = \frac{5}{13}$$

$$\Rightarrow \quad \theta = \sin^{-1} \frac{5}{13}$$

$$\text{Now to find } \sin^{-1} \frac{3}{5} - \sin^{-1} \frac{5}{13}$$

$$\text{We know } \sin^{-1} x - \sin^{-1} y = \sin^{-1} \left[x\sqrt{1-y^2} - y\sqrt{1-x^2} \right]$$

$$\begin{aligned}\text{LHS: } \sin^{-1} \frac{3}{5} - \sin^{-1} \frac{5}{13} &= \sin^{-1} \left[\frac{3}{5} \sqrt{1 - \left(\frac{5}{13}\right)^2} - \frac{5}{13} \sqrt{1 - \left(\frac{3}{5}\right)^2} \right] \\ &= \sin^{-1} \left[\frac{3}{5} \sqrt{\frac{144}{169}} - \frac{5}{13} \sqrt{\frac{16}{25}} \right] = \sin^{-1} \left[\frac{3}{5} \times \frac{12}{13} - \frac{5}{13} \times \frac{4}{5} \right] \\ &= \sin^{-1} \left[\frac{36}{65} - \frac{20}{65} \right] = \sin^{-1} \left(\frac{16}{65} \right)\end{aligned}$$

Question 5.

Prove that $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1} \left[\frac{x + y + z - xyz}{1 - xy - yz - zx} \right]$

Solution:

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x + y}{1 - xy} \right) = \tan^{-1} (A)$$

$$\text{Here } A = \frac{x + y}{1 - xy}$$

$$\text{So LHS: } \tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1} (A) + \tan^{-1} z$$

$$= \tan^{-1} \left(\frac{A + z}{1 - Az} \right) = \tan^{-1} \left[\frac{\frac{x + y}{1 - xy} + z}{1 - \frac{x + y}{1 - xy} (z)} \right] = \tan^{-1} \left[\frac{x + y + z(1 - xy)}{1 - xy - (x + y)z} \right]$$

$$= \tan^{-1} \left[\frac{x + y + z - xyz}{1 - xy - xz - yz} \right] = \text{RHS.}$$

Question 6.

If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$, show that $x + y + z = xyz$.

Solution:

Given $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$

$$\Rightarrow \tan^{-1} \left(\frac{x + y + z - xyz}{1 - xy - yz - zx} \right) = \pi$$

$$\Rightarrow \frac{x + y + z - xyz}{1 - xy - yz - zx} = \tan \pi = 0$$

$$\Rightarrow x + y + z - xyz = 0 \quad \Rightarrow \quad x + y + z = xyz$$

Question 7.

Prove that $\tan^{-1} x + \tan^{-1} \frac{2x}{1-x^2} = \tan^{-1} \frac{3x-x^3}{1-3x^2}$, $|x| < \frac{1}{\sqrt{3}}$.

Solution:

$$\begin{aligned} \text{LHS: } \tan^{-1} x + \tan^{-1} \frac{2x}{1-x^2} &= \tan^{-1} \left[\frac{x + \frac{2x}{1-x^2}}{1 - x \left(\frac{2x}{1-x^2} \right)} \right] = \tan^{-1} \left[\frac{\frac{x(1-x^2) + 2x}{1-x^2}}{1 - \frac{x(2x)}{1-x^2}} \right] \\ &= \tan^{-1} \left[\frac{\frac{x - x^3 + 2x}{1-x^2}}{\frac{1-x^2-2x^2}{1-x^2}} \right] = \tan^{-1} \left[\frac{x - x^3 + 2x}{\frac{1-3x^2}{1-x^2}} \right] = \tan^{-1} \frac{(x - x^3 + 2x)(1-x^2)}{1-3x^2} \\ &= \tan^{-1} \frac{3x - x^3}{1-3x^2} \\ &= \text{RHS} \end{aligned}$$

Question 8.

Simplify: $\tan^{-1} \frac{x-y}{y} - \tan^{-1} \frac{x-y}{x+y}$

Solution:

$$\tan^{-1} \frac{x-y}{y} - \tan^{-1} \frac{x-y}{x+y} = \tan^{-1} \left[\frac{\frac{x}{y} - \frac{y}{y}}{1 + \left(\frac{x}{y}\right)\left(\frac{x-y}{x+y}\right)} \right] = \tan^{-1} \left[\frac{\frac{x(x+y) - y(x-y)}{y(x+y)}}{\frac{y(x+y) + x(x-y)}{y(x+y)}} \right]$$

$$= \tan^{-1} \left[\frac{\frac{x^2 + xy - xy + y^2}{y(x+y)}}{\frac{y(x+y) + x(x-y)}{y(x+y)}} \right] = \tan^{-1} \left[\frac{x^2 + y^2}{y(x+y) + x^2 - xy} \right]$$

$$= \tan^{-1} \left(\frac{x^2 + y^2}{xy + y^2 + x^2 - xy} \right) = \tan^{-1} \left(\frac{x^2 + y^2}{x^2 + y^2 + xy} \right)$$

$$= \tan^{-1} \left(\frac{x^2 + y^2}{x^2 + y^2} \right) = \tan^{-1} (1) = \frac{\pi}{4}$$

Question 9.

Find the value of

(i) $\sin^{-1} \frac{5}{x} + \sin^{-1} \frac{12}{x} = \frac{\pi}{2}$

(ii) $2 \tan^{-1} x = \cos^{-1} \frac{1-a^2}{1+a^2} - \cos^{-1} \frac{1-b^2}{1+b^2}$, $a > 0$, $b > 0$

(iii) $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$

(iv) $\cot^{-1} x - \cot^{-1}(x+2) = \frac{\pi}{12}$, $x > 0$

Solution:

(i) Given $\sin^{-1} \frac{5}{x} + \sin^{-1} \frac{12}{x} = \frac{\pi}{2}$

$$\Rightarrow \sin^{-1} \frac{5}{x} = \frac{\pi}{2} - \sin^{-1} \frac{12}{x} = \cos^{-1} \frac{12}{x} \quad \left(\because \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \right)$$

so now $\sin^{-1} \frac{5}{x} = \cos^{-1} \frac{12}{x}$

Let $\sin^{-1} \frac{5}{x} = \theta$

$$\Rightarrow \sin \theta = \frac{5}{x}$$

But we know $\sin^2 \theta + \cos^2 \theta = 1$

$$\Rightarrow \left(\frac{5}{x} \right)^2 + \left(\frac{12}{x} \right)^2 = 1 \quad \text{i.e.} \quad \frac{25}{x^2} + \frac{144}{x^2} = 1 \quad \Rightarrow \quad x^2 = 169$$

$$a = 13$$

(ii) Let $2 \tan^{-1} x = \cos^{-1} \frac{1-a^2}{1+a^2} - \cos^{-1} \frac{1-b^2}{1+b^2}$

Put $a = \tan \theta_1 \Rightarrow \theta_1 = \tan^{-1} a$

Now $\frac{1-a^2}{1+a^2} = \frac{1-\tan^2 \theta_1}{1+\tan^2 \theta_1} = \cos 2\theta_1$

Similarly, put $b = \tan \theta_2 \Rightarrow \theta_2 = \tan^{-1} b$

Now $\frac{1-b^2}{1+b^2} = \cos 2\theta_2$

so $\cos^{-1} \left(\frac{1-a^2}{1+a^2} \right) = \cos^{-1}(\cos 2\theta_1) = 2\theta_1 = 2 \tan^{-1} a$

and $\cos^{-1} \left(\frac{1-b^2}{1+b^2} \right) = \cos^{-1}(\cos 2\theta_2) = 2\theta_2 = 2 \tan^{-1} b$

Now $2 \tan^{-1} x = 2 \tan^{-1} a - 2 \tan^{-1} b$

$$\Rightarrow (\div \text{ by } 2) \quad \tan^{-1} x = \tan^{-1} a - \tan^{-1} b$$

i.e. $\tan^{-1} x = \tan^{-1} \left(\frac{a-b}{1+ab} \right) \Rightarrow x = \frac{a-b}{1+ab}$

(iii) $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$

$$\begin{aligned} \text{LHS} &= 2 \tan^{-1}(\cos x) = \tan^{-1}(\cos x) + \tan^{-1}(\cos x) \\ &= \tan^{-1} \left(\frac{\cos x + \cos x}{1 - \cos^2 x} \right) = \tan^{-1} \left(\frac{2 \cos x}{\sin^2 x} \right) \end{aligned}$$

RHS = $\tan^{-1}(2 \operatorname{cosec} x)$

Now LHS = RHS

$$\Rightarrow \tan^{-1} \left(\frac{2 \cos x}{\sin^2 x} \right) = \tan^{-1}(2 \operatorname{cosec} x)$$

$$\Rightarrow \frac{2 \cos x}{\sin^2 x} = 2 \operatorname{cosec} x = \frac{2}{\sin x}$$

$$2 \cot x = 2$$

$$\Rightarrow \cot x = 1 \Rightarrow x = \frac{\pi}{4}$$

(iv) $\cot^{-1} x - \cot^{-1}(x+2) = \frac{\pi}{12}$, $x > 0$

$$= \tan^{-1} \left[\frac{\frac{1}{x} - \frac{1}{x+2}}{1 + \left(\frac{1}{x} \right) \left(\frac{1}{x+2} \right)} \right] = \tan^{-1} \left[\frac{x+2-x}{x(x+2)+1} \right] \cdot \frac{x(x+2)+1}{x(x+2)}$$

$$= \tan^{-1} \left(\frac{2}{x^2+2x+1} \right) = \frac{\pi}{12}$$

$$\Rightarrow \frac{2}{x^2+2x+1} = \tan \frac{\pi}{12} = \tan 15^\circ = \tan(45^\circ - 30^\circ)$$

$$= \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ} = \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}}$$

$$\Rightarrow \frac{2}{x^2+2x+1} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

$$\Rightarrow 2(\sqrt{3}+1) = (\sqrt{3}-1)(x^2+2x+1)$$

$$\Rightarrow x^2+2x+1 = \frac{2(\sqrt{3}+1)}{\sqrt{3}-1} = \frac{2(\sqrt{3}+1)(\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)} = \frac{3+2\sqrt{3}+2\sqrt{3}+1}{3-1} = 2+\sqrt{3}$$

$$\Rightarrow (x+1)^2 = \sqrt{3}+3 \Rightarrow x+1 = \sqrt{\sqrt{3}+3} \Rightarrow x = \sqrt{\sqrt{3}+3} - 1$$

Question 10.

Find the number of solution of the equation $\tan^{-1}(x-1) + \tan^{-1}x + \tan^{-1}(x+1) = \tan^{-1}(3x)$.

Solution:

$$\tan^{-1}(x-1) + \tan^{-1}x + \tan^{-1}(x+1) = \tan^{-1}(3x)$$

$$\tan^{-1}(x-1) + \tan^{-1}(x+1) = \tan^{-1}3x - \tan^{-1}x$$

$$\text{LHS: } \tan^{-1} \frac{(x-1)+(x+1)}{1-(x-1)(x+1)} = \tan^{-1} \frac{2x}{1-(x^2-1)} = \tan^{-1} \frac{2x}{2-x^2} \quad \dots\dots (1)$$

$$\text{RHS: } \tan^{-1}3x - \tan^{-1}x = \tan^{-1} \frac{3x-x}{1+(3x)(x)} = \tan^{-1} \frac{2x}{1+3x^2} \quad \dots\dots (2)$$

$$\text{LHS} = \text{RHS}$$

$$\Rightarrow \tan^{-1} \frac{2x}{2-x^2} = \tan^{-1} \frac{2x}{1+3x^2}$$

$$\Rightarrow \frac{2x}{2-x^2} = \frac{2x}{1+3x^2}$$

$$\Rightarrow 2-x^2 = 1+3x^2$$

$$\Rightarrow 4x^2 = 1$$

$$\Rightarrow x^2 = \frac{1}{4}$$

$$\Rightarrow x = \pm \frac{1}{2}$$

So, the equation has 2 solutions.